

Brandon Kong

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SKILLS

Coding Languages Python, TypeScript, JavaScript, C++, SQL, HTML, CSS
Frameworks Next, React, Express, Langchain, TensorFlow, PyTorch, OpenCV, Scikit-learn
Tools OpenAI, AWS, Docker, Prometheus, Grafana, MongoDB, Supabase, Git, Bash, WSL, VS Code

WORK EXPERIENCE

Rogers Communications Sept 2025 - Present
Machine Learning Intern | *Python, AWS* Toronto, ON

- Partnered with **Amazon** to prototype a centralized Roger's services app using **AWS Bedrock** that delivers hyper-personalized bundle offers to boost customer adoption rates, projected to increase revenue by **\$205M–310M**.
- Designed a multi-agent orchestration platform using **AWS Bedrock, Lambda, and DynamoDB** that automates end-to-end Buyflows, including identity verification, background checks, and real-time credit pulls.
- Optimized insights across 20,000+ employee survey complaints by building a **ETL+AI** sentiment-analysis pipeline, including ingestion, cleaning, theme extraction, and a chatbot interface to surface insights instantly.

Kisoji Biotechnology May 2025 - Sept 2025
Machine Learning Engineer | *Python, Pytorch, FastAPI* Toronto, ON

- Work on AI-enabled generative models to analyze antibody sequence data, support paratope fingerprinting, and assist in lead candidate optimization within KisoJi's proprietary antibody discovery platform.
- Trained models to predict binding activity of AI-generated antibodies, saving over **\$20,000** in lab costs.

Adanomad Jan 2025 - Apr 2025
AI Full Stack Software Engineer | *Python, TypeScript, OpenAI, Next.js, Supabase* Hamilton, ON

- Built a real-time call handling interface using WebSockets and concurrent request pooling, supporting 200+ users.
- Designed a modular **RAG+LLM** pipeline to automate agentic workflows, cutting response latency by 80%.
- Integrated dynamic toolchains via LangChain to enable AI agents to interact with external APIs like DALL·E and Gemini.

UW Visual Image Processing Lab May 2024 - Aug 2024
Software Developer | *Python, C++, CLI, .NET* Waterloo, ON

- Improved region-growing segmentation with edge-penalty tuning, increasing classification accuracy by **20%**.
- Developed Python-based preprocessing pipelines, boosting image analysis throughput by **200%**.

PROJECTS

ClassiMail 📧 | *Python, Flask, Next, OpenAI, SQLite*

- Designed and deployed a full-stack Gmail classification pipeline to process and categorize over **5,000+ emails**, leveraging **OpenAI** models for real-time filtering and sender extraction.
- Integrated **Prometheus, Grafana, and Docker**, enabling real-time metrics and performance tracking.
- Leveraged vectorized text and image embeddings to expand the knowledge base and enhance email classification accuracy.

DigiClassifier | *Python, NumPy*

- Built a fully functional customized **FCN** with PyTorch, capable of classifying MNIST digits with high accuracy
- Implemented a robust training loop with **BCE** loss, Adam optimizer and gradient clipping, monitoring accuracy.

JARVIS | *Python, SQLite*

- Developed and deployed a Python-based **AI agent** with context-aware learning, built on an end-to-end **RAG** pipeline handling input preprocessing, context retrieval, and adaptive response generation.
- Planned system flow and failover mechanisms to ensure robustness across concurrent queries, edge-case inputs, and **ACID-compliant transactions**.

EDUCATION

Honours Bachelor of Computer Engineering 2028

- Relevant Courses: Digital Computers, Linear Algebra, Algorithms and Data Structures, Compilers